

Demo 1 (a) : Original TheSky scripting logistics test

```
sky6StarChart.Find("Jupiter");
```

```
sky6ObjectInformation.Property(59);
```

```
var alt = sky6ObjectInformation.ObjInfoPropOut;
```

```
sky6ObjectInformation.Property(58);
```

```
var az = sky6ObjectInformation.ObjInfoPropOut;
```

```
sky6ObjectInformation.Property(65);
```

```
var mag = sky6ObjectInformation.ObjInfoPropOut;
```

```
sky6ObjectInformation.Property(70);
```

```
var ha = sky6ObjectInformation.ObjInfoPropOut;
```

```
RunJavaScriptOutput =
```

```
"Altitude: " + alt + "\n" +
```

```
"Azimuth: " + az + "\n" +
```

```
"Magnitude: " + mag + "\n" +
```

```
"Hour Angle: " + ha;
```

(b) TheSky scripting with SkyWrapper

```
sky6StarChart.Find("Jupiter");
```

```
var pos = SkyWrapper.Position.getAllPosition();  
var mag = SkyWrapper.Physical.getMagnitude();
```

```
RunJavaScriptOutput =  
  "Altitude: " + pos.altitude + "\n" +  
  "Azimuth: " + pos.azimuth + "\n" +  
  "Magnitude: " + mag + "\n" +  
  "Hour Angle: " + pos.hourAngle;
```

Demo 2: Full system environment report

Should yield: Full environment report, recognizing what system TheSky X is running on, the build, and version numbers listed in an array.

```
RunJavaScriptOutput = SkyWrapper.System.getEnvironmentReport();
```

Demo 3: Full Profile Test

Should yield: Queried object, and all its properties as listed in the query()

```
var d = SkyWrapper.query("Jupiter");  
  
var lines = [];  
lines.push("=====");  
lines.push(" SKYWRAPPER FULL PROFILE");  
lines.push("=====");
```

```
lines.push("");
lines.push(" Object   : " + d.name);
lines.push(" Queried At : " + d.queriedAt);
lines.push(" Type     : " + d.identity.typeName);
lines.push(" Primary Name: " + d.identity.primaryName);
lines.push(" Observable : " + (d.observable ? "YES" : "NO"));
lines.push("");
lines.push(" POSITION");
lines.push(" -----");
lines.push(" Altitude  : " + d.position.altitude.toFixed(2) + "°");
lines.push(" Azimuth   : " + d.position.azimuth.toFixed(2) + "°");
lines.push(" Hour Angle : " + d.position.hourAngle.toFixed(4) + " hrs");
lines.push(" Air Mass   : " + d.position.airMass.toFixed(3));
lines.push("");
lines.push(" COORDINATES");
lines.push(" -----");
lines.push(" RA (now)   : " + d.coordinates.raString);
lines.push(" Dec (now)  : " + d.coordinates.decString);
lines.push(" RA (J2000) : " + d.coordinates.ra2000String);
lines.push(" Dec (J2000) : " + d.coordinates.dec2000String);
lines.push("");
lines.push(" PHYSICAL");
lines.push(" -----");
lines.push(" Magnitude  : " + d.physical.magnitude.toFixed(2));
lines.push(" Phase      : " + d.physical.phase.toFixed(1) + "%");
lines.push(" Mag Reliable: " + (d.physical.magnitudeReliable ? "YES" : "NO"));
```

```
lines.push("");  
lines.push(" TIMING SOURCE: " + d.timingSource.toUpperCase());  
lines.push(" " + d.timing);  
lines.push("");  
lines.push("=====");
```

```
RunJavaScriptOutput = lines.join("\n");
```

Demo 4: Arbitrary Observing list status test

Should yield: Queries 10 objects, classifies them as observable or not observable, (“Good” or “Bad”) then outlines their properties.

```
var targets = [  
    "Jupiter",  
    "Saturn",  
    "Mars",  
    "Vega",  
    "Arcturus",  
    "M13",  
    "M57",  
    "M5",  
    "Antares",  
    "M4"  
];
```

```
var good = [];
```

```
var low = [];
```

```
var failed = [];
```

```
for (var i = 0; i < targets.length; i++) {
```

```
    var d = SkyWrapper.query(targets[i]);
```

```
    if (d === null) {
```

```
        failed.push(targets[i]);
```

```
        continue;
```

```
    }
```

```
var mag = (d.physical.magnitude !== null)
```

```
    ? d.physical.magnitude.toFixed(2)
```

```
    : "N/A";
```

```
var timing = "N/A";
```

```
if (d.timing !== null) {
```

```
    if (d.timingSource === "solar_system") {
```

```
        timing = "see solar system times";
```

```
    } else {
```

```
        timing = "transit " + d.timing.transitString;
```

```
    }
```

```
}
```

```

var entry =

    d.name + " | " +

    d.identity.typeName + " | " +

    "Alt: " + d.position.altitude.toFixed(1) + "° | " +

    "Az: " + d.position.azimuth.toFixed(1) + "° | " +

    "Mag: " + mag + " | " +

    timing;

if (d.observable) {

    good.push(entry);

} else {

    low.push(entry);

}

}

var lines = [];

lines.push("=====");

lines.push(" TONIGHT'S OBSERVING LIST");

lines.push(" " + SkyWrapper.Time.getCurrentDateString() + " UTC");

lines.push(" Sidereal: " + SkyWrapper.Time.getLocalSiderealTimeString());

lines.push("=====");

lines.push("");

lines.push(" OBSERVABLE (" + good.length + " targets)");

lines.push(" " + new Array(44).join("-"));

for (var g = 0; g < good.length; g++) {

```

```
    lines.push(" ✓ " + good[g]);  
}
```

```
lines.push("");  
lines.push(" BELOW THRESHOLD (" + low.length + " targets)");  
lines.push(" " + new Array(44).join("-"));
```

```
for (var l = 0; l < low.length; l++) {  
    lines.push(" X " + low[l]);  
}
```

```
if (failed.length > 0) {  
    lines.push("");  
    lines.push(" NOT FOUND: " + failed.join(", "));  
}
```

```
lines.push("");  
lines.push("=====");
```

```
RunJavaScriptOutput = lines.join("\n");
```

Demo 5: Object Type Intelligence

```

var targets = ["Sirius", "Jupiter", "M42", "M31", "M13", "Moon"];

var lines = [];

lines.push("=====");
lines.push(" CROSS-OBJECT-TYPE INTELLIGENCE TEST");
lines.push("=====");
lines.push("");

for (var i = 0; i < targets.length; i++) {
    var d = SkyWrapper.query(targets[i]);
    if (d === null) { lines.push(" FAILED: " + targets[i]); continue; }

    var mag = d.physical.magnitude !== null
        ? d.physical.magnitude.toFixed(2) +
          (d.physical.magnitudeReliable ? "" : " [unreliable]")
        : "N/A";

    lines.push(" " + d.name);
    lines.push("  Type   : " + d.identity.typeName);
    lines.push("  Alt/Az  : " + d.position.altitude.toFixed(1) +
              "° / " + d.position.azimuth.toFixed(1) + "°");
    lines.push("  Magnitude: " + mag);
    lines.push("  Timing  : " + d.timingSource);
    lines.push("  Observable: " + (d.observable ? "YES" : "NO"));

    if (d.identity.spectralClass !== null) {

```

```

        lines.push("  Spectral : " + d.identity.spectralClass);
    }
    if (d.physical.phase !== null) {
        lines.push("  Phase   : " + d.physical.phase.toFixed(1) + "%");
    }
    if (d.physical.majorAxis !== null) {
        lines.push("  Size    : " + d.physical.majorAxis.toFixed(1) +
            " x " + d.physical.minorAxis.toFixed(1) + " arcmin");
    }

    lines.push("");
}

```

```
RunJavaScriptOutput = lines.join("\n");
```

Demo 6: Identify Module & Catalog Depth

```
var targets = ["Sirius", "M42", "M13"];
```

```
var lines = [];
```

```
lines.push("=====");
```

```
lines.push(" CATALOG IDENTITY REPORT");
```

```
lines.push("=====");
```

```
for (var i = 0; i < targets.length; i++) {
```

```
sky6StarChart.Find(targets[i]);  
lines.push("");  
lines.push(SkyWrapper.Identity.getAllNames());  
}
```

```
RunJavaScriptOutput = lines.join("\n");
```

Demo 7: Planning data and separation

```
var lines = [];
```

```
// Time report
```

```
lines.push("=====");  
lines.push(" SESSION PLANNING REPORT");  
lines.push("=====");  
lines.push("");  
lines.push(SkyWrapper.Time.getTimeReport());  
lines.push("");
```

```
// Moon position and phase
```

```
var moon = SkyWrapper.query("Moon");  
if (moon !== null) {  
    lines.push(" MOON STATUS");  
    lines.push(" " + new Array(34).join("-"));
```

```

lines.push(" Altitude : " + moon.position.altitude.toFixed(1) + "°");
lines.push(" Azimuth  : " + moon.position.azimuth.toFixed(1) + "°");
lines.push(" Phase    : " + moon.physical.phase.toFixed(1) + "%");
lines.push(" Above threshold: " +

    (moon.observable ? "YES — consider moon avoidance" : "NO — moon below
horizon"));

lines.push("");
}

// Angular separation from moon for key targets
var targets = ["Jupiter", "M13", "Vega", "Antares"];

lines.push(" MOON SEPARATION");
lines.push(" " + new Array(34).join("-"));

for (var i = 0; i < targets.length; i++) {
    var sep = SkyWrapper.Extras.getAngularSeparation(targets[i], "Moon");
    if (sep !== null) {
        var warning = sep < 15 ? " ← AVOID" : sep < 30 ? " ← CAUTION" : "";
        lines.push(" " + targets[i] +
            ": " + sep.toFixed(1) + "°" + warning);
    }
}

lines.push("");

```

```
// Refraction check for a low target
sky6StarChart.Find("Antares");
sky6ObjectInformation.Property(59);
var antaresAlt = sky6ObjectInformation.ObjInfoPropOut;
var refraction = SkyWrapper.Extras.getRefraction(antaresAlt);

lines.push(" REFRACTION");
lines.push(" " + new Array(34).join("-"));
lines.push(" Antares altitude : " + antaresAlt.toFixed(1) + "°");
lines.push(" Refraction      : " + refraction.toFixed(2) + " arcsec");
lines.push("");
lines.push("=====");

RunJavaScriptOutput = lines.join("\n");
```